

Name		Index No.		Civics Group	
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DUNMAN HIGH SCHOOL
SENIOR HIGH
2008 PRELIMINARY EXAMINATIONS
General Certificate of Education Advanced Level
YEAR SIX H2 BIOLOGY (9747)

BIOLOGY

9747 / 03

PAPER 3 Applications

22 September 08 (Monday)

1 hour 30 minutes

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST.

Write your name, civics group and index number on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Do not open this booklet until you are told to do so.

Section D:

Consists of 4 Structured Questions

Answer **all** questions.

Write your answers in the **space provided** on the question paper.

Section E:

Consists of 1 Essay Questions

Answer **all** questions.

Write your answers on the separate **writing papers** provided.

At the end of the examination, fasten all your work securely together. The number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE		
Section D	1	
	2	
	3	
	4	
Section E	5	
Total		

This question paper consists of **12** printed pages and **2** blank pages (including this cover page).

Section D: Structured Questions (80 marks)Answer **all** questions in this section.**Question 1**

The coat colour of Norwegian cattle is mainly determined by the distribution of two pigments: red and black. Both pigments are produced by the action of the enzyme tyrosinase in cells called melanocytes. A low level of activity of the enzyme leads to the production of red pigment, whilst a high activity allows only black pigment production. The activity of the enzyme is increased by melanocyte stimulating hormone (MSH), which combines with an MSH receptor. The receptor is coded for by the E locus, which has three alleles, E^D , E^A and e . E^D and E^A each give a receptor with a different activity. No receptor is produced by the recessive allele, e .

<i>E locus</i>		<i>A locus</i>	
<i>genotype</i>	<i>MSH receptor</i>	<i>genotype</i>	<i>protein which blocks MSH receptor</i>
$E^D E^D$ or $E^D e$	insensitive to A locus blocking protein	AA or Aa	present
$E^A E^A$ or $E^A e$	sensitive to A locus blocking protein	aa	absent
ee	none		

Allele E^A differs from E^D by a single base substitution and e differs from E^A by single base deletion.

- (a) Suggest how these mutations might result in differences in the MSH receptor.
[2]

DNA was extracted from the frozen semen of six bulls with different genotypes at the E locus. The DNA from each animal was separately digested with two different restriction enzymes, P and Q. The products of each digestion were separated on a gel. The banding patterns produced with respect to this locus are shown in **Figure 1.1**

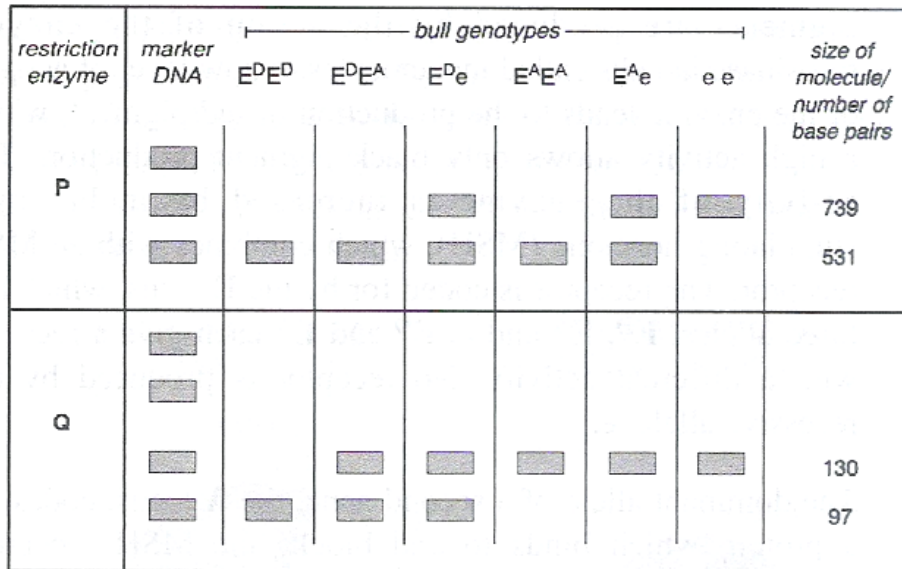


Fig. 1.1

(b) Explain briefly

- (i)** how the products of digestion of DNA with restriction enzymes can be separated on a gel; [2]

- (ii)** why the products of digestion of DNA from the same animal are different when a different restriction enzyme is used; [2]

(iii) the role of marker DNA. [2]

(c) State which genotype can be identified by using each of the two restriction enzymes. [2]

P

Q

Total: [10]

Question 2

Figure 2 shows the main stages in the polymerase chain reaction (PCR).

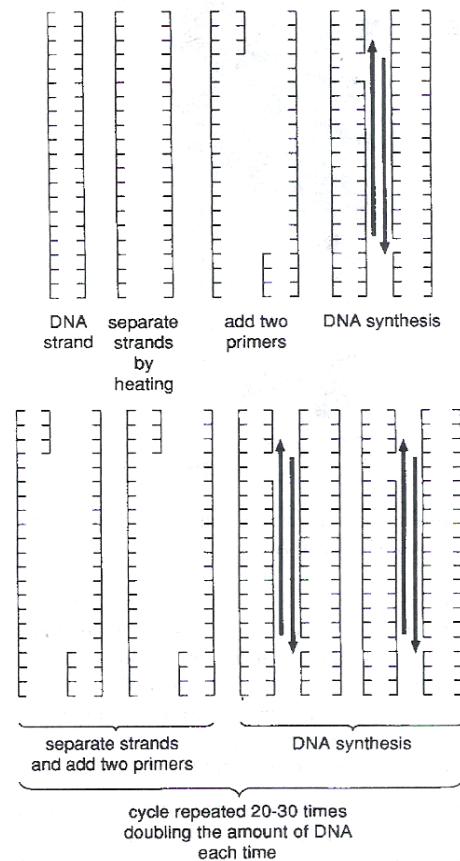


Figure 2

This technique allows multiple copies of a selected length of DNA to be made in the laboratory. Portions of the base sequences at either end of the DNA to be copied are used to produce two short lengths of synthetic DNA, one complementary to each strand of the DNA double helix. These short lengths of DNA serve as primers for *in vitro* DNA synthesis, which is catalysed by DNA polymerase, and they determine the ends of the final DNA fragment that is obtained.

Each cycle of the reaction requires a brief heat treatment to separate the two strands of the DNA double helix. A subsequent cooling of the DNA in the presence of a large excess of the two DNA primers allows their specific attachment to their complementary DNA polymerase and the **four different DNA nucleotides** so that the regions of DNA, starting from the primers, are selectively synthesised.

When first developed, PCR was slow and laborious because a crucial enzyme for the process, DNA polymerase, was denatured at high temperature and had to be added to each cycle. However the introduction of the thermostable enzyme, *Taq*

DNA polymerase (isolated from the heat tolerant bacterium, *Thermus aquaticus*) revolutionised PCR so that the technique could be fully automated.

(a) (i) List the components of a nucleotide. [2]

(ii) Explain what is meant by **four different DNA nucleotides**. [2]

(b) Suggest why the brief heat treatment separates the two strands of the DNA double helix. [2]

(c) Suggest why DNA synthesis takes place in opposite directions on the two separated strands of the double helix. [2]

(d) Explain the advantages of using *Taq* DNA polymerase. [2]

Total: [10]

Question 3

(a) Explain what is meant by gene therapy. [10]

(b) Describe the possible benefits of gene therapy for humans. [5]

(c) Discuss the ethical implications of genetic engineering in humans. [5]

[Total: 20]

Question 4

The first British trial of a possible gene therapy for cystic fibrosis (CF) began in 1993. Nine adult male patients with advanced lung disease were chosen. Six patients received gene therapy and three acted as a control.

The normal allele of the CF gene codes for a protein that forms an ion channel in the membrane of the epithelial cells that line the gut and the air-ways. In sufferers of CF, chloride ions cannot pass through this channel.

The purpose of the trial was to see whether functioning ion channels could be manufactured by cells which have had the normal allele of the CF gene inserted into them. DNA coding for the normal allele was cloned and put into small spheres of phospholipid bilayers (liposomes). The droplets were inhaled as an aerosol into the nose. Liposomes were thought to be a safer vector than the adenoviruses used in the earlier trials in North America.

Gene therapy cannot provide a permanent cure for cystic fibrosis since cells from the airways are shed constantly.

Male patients were chosen for the British trial to avoid the risks of the allele being passed into the germ-line, since virtually all men with CF are infertile.

(a) Explain how cystic fibrosis is inherited. [3]

(b) (i) Suggest how a liposome can be a suitable vector for introducing DNA into mammalian cells. [2]

- (ii) Outline one method by which DNA coding for the normal CF allele could have been isolated and cloned for this therapy. [3]

- (iii) Comment on **one** possible hazard of this example of genetic engineering. [2]

[Total: 10]

Section E: Free-response Questions (20 marks)

Answer **all** questions.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, wherever appropriate.

Your answer must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 5** **(a)** Discuss the advantages and disadvantages of cloning. [6]
- (b)** Describe one example of the use of genetic engineering in agriculture. [6]
- (c)** Outline the use in DNA analysis of
- (i)** Restriction enzymes; [4]
- (ii)** Electrophoresis. [4]

Total: [20]